

# Networking Legacy Services & Web Standards

## Lesson 53 — Classic Services and Standards

JKSSB Computer Awareness

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# Why This Lesson Is a Trap-Drill

- ▶ Every item in this lesson tests a **near-neighbour** pair: two terms that look alike but do different jobs.
- ▶ The exam's favourite weapon is the **negative stem**: NOT, EXCEPT, INCORRECT statement.
- ▶ Read every option; the wrong answers are built from **real traps** in the trap registry, not random words.
- ▶ Rule: match the **service to its role**, the **term to its full form**, and the **statement to its defect**.

## How to survive

Slow down on NOT/EXCEPT items. The correct answer is the one that breaks the pattern.

# Packet Switching vs Circuit Switching

- ▶ **Circuit switching** sets up a dedicated path for the whole call before any data moves; the classic example is the old telephone call.
- ▶ **Packet switching** breaks data into packets; each packet can travel independently and the path is **not** reserved in advance.
- ▶ The Internet runs on **packet switching**, not on a dedicated circuit per session.

## Keep the pair straight

Dedicated reserved path = circuit switching; independent packets, no reserved path = packet switching.

## Telnet: Remote Login

- ▶ **Telnet** — TELecommunication NETwork — gives remote login to another computer over the network.
- ▶ It lets a user run commands on the **remote** machine as if sitting at its keyboard.
- ▶ Telnet sends data, including passwords, in **plain text**; it is **not** a file-transfer or mail service.

### Trap alert

Telnet = remote login. It is not FTP (file transfer) and not SMTP (mail sending).

# FTP: File Transfer

- ▶ **FTP** — File Transfer Protocol — transfers files between computers on a network.
- ▶ It is the service used to **upload and download files**, not to log in and run commands, and not to send mail.
- ▶ Keep the triple apart: **Telnet** logs in remotely, **FTP** moves files, **SMTP** sends mail.

## Service-role match

Remote login = Telnet; file transfer = FTP; mail sending = SMTP.

# SMTP, POP3, IMAP: Who Does What

- ▶ **SMTP** — Simple Mail Transfer Protocol — **sends** mail from the sender toward the receiver's mail server.
- ▶ **POP3** — Post Office Protocol version 3 — **downloads** mail to the local machine, typically removing it from the server.
- ▶ **IMAP** — Internet Message Access Protocol — keeps mail **on the server** and syncs it across devices.

## Trap alert

POP3 downloads to the local machine; it does not keep mail synced on the server. That synced behaviour is IMAP. (TRAP-17)

# Mail Traps: Signature, Encryption, BCC

- ▶ A **digital signature** proves authenticity and integrity; it does **not** encrypt the body. Only **encryption** gives confidentiality. (TRAP-15)
- ▶ **BCC** hides recipients from **each other**; the sender always knows, and To/Cc names stay visible to all. (TRAP-16)
- ▶ In an address, **@** separates the **user name** from the **domain name**. (TRAP-28)

## Three one-line defences

Signature = who sent it unchanged; encryption = only the recipient reads it; BCC = recipients hidden from each other.

## W3C: Who Sets Web Standards

- ▶ **W3C** — World Wide Web Consortium — develops the open standards of the World Wide Web such as HTML and CSS.
- ▶ It is a **standards body**, not a browser, not a search engine, and not an Internet service provider.
- ▶ Near-neighbour guard: **ISP** — Internet Service Provider — gives Internet access; W3C writes the standards.

Keep the pair straight

W3C = web standards; ISP = Internet access. Do not swap them.

# DHTML vs HTML

- ▶ **HTML** — HyperText Markup Language — marks up the static structure of a web page.
- ▶ **DHTML** — Dynamic HyperText Markup Language — is **not** a separate language; it is the combination of HTML with styling and scripting that makes pages dynamic.
- ▶ Trap: treating DHTML as a brand-new replacement language for HTML is wrong; it **builds on** HTML.

## Office desktop example

A static notice page uses HTML; the same page with a drop-down menu that reacts to clicks uses DHTML techniques on top of that HTML.

# URL, DNS, IP Address

- ▶ A **URL** — Uniform Resource Locator — is the full address of a resource on the web.
- ▶ **DNS** — Domain Name System — translates human-readable domain names into numeric addresses.
- ▶ An **IP address** is the numeric address of a device on the network; DNS resolves names **to** IP addresses.

## Chain in one line

Type a URL, DNS resolves its domain name, the browser reaches the IP address.

## IPv4 vs IPv6

- ▶ **IPv4** uses **32-bit** addresses, written as four decimal groups.
- ▶ **IPv6** uses **128-bit** addresses — never 256-bit — with a fixed header of **40 bytes**. (TRAP-18)
- ▶ IPv6 was introduced because the IPv4 address space is too small; the bit-length jump is 32 to 128.

### Trap alert

Any option saying IPv6 is 256-bit is a planted distractor from a real paper (JA 2026 Q78). Reject it on sight.

# The Seven OSI Layers

- ▶ The **OSI model** has **seven** layers, top to bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.
- ▶ Memory hook: **All People Seem To Need Data Processing**.
- ▶ Near-neighbour guard: the **TCP/IP model** has four layers, not seven — do not answer “four” when the stem says OSI.

## The one number

OSI = seven layers, always. Four layers = TCP/IP model, a different question.

# Physical vs Logical Ports

- ▶ A **physical port** is a hardware socket on the machine where a cable or device plugs in.
- ▶ A **logical port** is a number that identifies a service or process in networking software.
- ▶ Trap: a logical port number is **not** a hardware socket, and a physical socket is **not** a service number.

Keep the pair straight

Hardware socket = physical port; service number = logical port.

## Companion Protocol Traps

- ▶ **ICMP** is a control and diagnostic protocol; it transfers **no** application data and uses no ports. (TRAP-19)
- ▶ **BGP** routes **between** autonomous systems (inter-domain); it does not route inside one local network. (TRAP-20)
- ▶ **HTTP/3** runs over **QUIC over UDP**, not over TCP. (TRAP-21)

### Trap alert

“ICMP transfers data”, “BGP is for one LAN”, “HTTP/3 uses TCP” — all three are planted wrong statements. Reject each.

## Trap Alert: Patterns to Reject on Sight

- ▶ POP3 keeps mail synced on the server — wrong; that is IMAP. (TRAP-17)
- ▶ Digital signature encrypts the body — wrong; encryption gives confidentiality. (TRAP-15)
- ▶ IPv6 is 256-bit — wrong; it is 128-bit. (TRAP-18)
- ▶ Telnet transfers files; FTP sends mail — wrong; Telnet logs in, FTP moves files, SMTP sends mail.
- ▶ OSI has four layers — wrong for OSI; seven is the OSI number.

## Lesson 53: Recall

- ▶ Circuit switching reserves a dedicated path; packet switching sends independent packets with no reserved path.
- ▶ Telnet = remote login; FTP = file transfer; SMTP sends mail, POP3 downloads it locally, IMAP syncs it on the server.
- ▶ W3C writes web standards; DHTML builds dynamic behaviour on top of HTML; URL names the resource, DNS resolves names, IP addresses the device.
- ▶ IPv4 is 32-bit; IPv6 is 128-bit with a 40-byte header.
- ▶ OSI has seven layers; physical ports are hardware sockets, logical ports are service numbers.
- ▶ ICMP carries no application data; BGP is inter-domain; HTTP/3 uses QUIC over UDP.